Spring Semester 2025–2026

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 1

To be covered during the examples class at 16:00 on Monday 9th February. These exercises do not count toward your final module mark, however it is a good idea to attempt the questions prior to the examples class.

1. Suppose that X_t is a variable from a time series at time t. Apply the following operators to X_t:

   (a) $B - B^3 + B^5$

   (b) $\psi(B) = 1 - \frac{1}{2}B + \frac{1}{4}B^2 - \frac{1}{6}B^4$

   (c) $\nabla(1 + B - B^2)$

2. Write the following process in $MA(\infty)$ form

$$\left(1 - \frac{1}{3}B\right) X_t = Z_t.$$

3. A mean zero stationary process $\{X_t\}$ has autocovariance function $\gamma_X(h)$. Suppose that $\{Y_t\}$ is a new process, defined

$$Y_t = X_t - X_{t-1}.$$

   (a) Show that $\{Y_t\}$ has mean zero for all t.

   (b) Determine the autocovariance function of $\{Y_t\}$ in terms of $\gamma_X(h)$.

4. Suppose that $\{Z_t\}$ is a white noise process with variance σ_z^2.

   (a) Show that the $MA(\infty)$ process, $\{X_t\}$, defined

$$X_t = Z_t + C(Z_{t-1} + Z_{t-2} + \ldots)$$

   where C is constant, is non-stationary where $C \neq 0$.

   (b) Show that the process $\{Y_t\}$, defined

$$Y_t = X_t - X_{t-1}$$

   is a stationary moving average process and hence determine the autocorrelation function of $\{Y_t\}$.

5. The process $\{Z_t\}$ is a white noise process with variance σ_z^2 and the process $\{X_t\}$ is defined

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}.$$

   (a) Show that $\mathbb{E}(X_t Z_{t-h}) = \sigma_z^2 \psi_h$.

   (b) Show that $\mathbb{E}(X_t X_{t-h}) = \sigma_z^2 \sum_{j=h}^{\infty} \psi_j \psi_{j-h}$.

6. Consider the following $MA(1)$ processes:

$$X_t = Z_t + \theta Z_{t-1}$$
$$Y_t = W_t + \frac{1}{\theta} W_{t-1}$$

where $\{Z_t\}$ and $\{W_t\}$ are mean zero white noise processes with variances σ^2 and $\theta^2\sigma^2$, respectively, with $|\theta| < 1$.

   (a) Show that $\{X_t\}$ and $\{Y_t\}$ have the same autocovariance function.

   (b) Determine the behaviour of $\{X_t\}$ and $\{Y_t\}$ for

      (i) $\theta \to 0$

      (ii) $\theta = 1$

7. The process $\{X_t\}$ is defined

$$X_t = U\cos(\omega t) + V\sin(\omega t) \quad t = 0, \pm 1, \pm 2, \ldots$$

where U and V are independent random variables, each with mean zero and variance 1, and $\omega \in [0, \pi]$ is a constant.

   (a) Show that the process $\{X_t\}$ is stationary and determine the mean, autocovariance and autocorrelation function for this process.

   (b) Suppose now that U and V are now correlated where $\mathrm{Cov}(U, V) = \xi \neq 0$ but that each of U and V still has mean zero and variance 1. Is the process $\{X_t\}$ still stationary? Justify your answer.

8. Consider the process

$$X_t = 5 + 3t + Z_t - \frac{1}{2} Z_{t-1}$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2.

   (a) Is this process stationary? Justify your answer.

(b) Show that the differenced process $Y_t = X_t - X_{t-1}$ is an $MA(2)$ process where

$$Y_t = \mu + Z_t + a_1 Z_{t-1} + a_2 Z_{t-2}$$

and state the values of μ, a_1 and a_2.

(c) Suppose that $a_2 = \frac{1}{2}$ but a_1 is unknown. Determine the range of values of a_1 for which the process $\{Y_t\}$ is invertible.